AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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24 marks · about 25 minutes

Key information

- Partial pressure = mole fraction × total pressure
- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets
- Find equilibrium amounts with a start / change / equilibrium table using the mole ratio in the equation
- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)
- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not

Section A: Short-answer questions

01 Nitrogen reacts with hydrogen in this exothermic reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Which change increases the equilibrium yield of ammonia but has no effect on the value of the equilibrium constant K_p?

[1 mark]

A Add a catalyst
B Increase the partial pressure of nitrogen
C Decrease the temperature
D Decrease the total pressure

02 State the effect, if any, of adding a catalyst on the value of K_c for an equilibrium.

[1 mark]

03 When HF is added to water at 298 K, this equilibrium is established. $HF(aq) \rightleftharpoons H^+(aq) + F^-(aq)$

At equilibrium, $[HF] = 7.70 \times 10^{-3}$ mol dm^{-3} and $[F^-] = 2.30 \times 10^{-3}$ mol dm^{-3}

What is the value of the equilibrium constant, in mol dm^{-3}, at 298 K?

[1 mark]

A 1.45×10^{3}
B 3.35
C 2.99×10^{-1}
D 6.87×10^{-4}

04 $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

The reaction reaches equilibrium in a container of fixed volume. Which is the expression for K_c for this equilibrium?

[1 mark]

A $K_c = [CH_3OH] \div ([CO] + [H_2]^2)$
B $K_c = [CH_3OH] \div ([CO][H_2]^2)$
C $K_c = ([CO] + [H_2]^2) \div [CH_3OH]$
D $K_c = [CO][H_2]^2 \div [CH_3OH]$

Section B: Longer questions

05 Methanol can be manufactured in a reversible reaction as shown.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \quad \Delta H^{\ominus} = -91 \text{ kJ mol}^{-1}$

05.1 A 0.230 mol sample of carbon monoxide is mixed with hydrogen in a 1:2 mol ratio and allowed to reach equilibrium in a sealed flask at temperature T. At equilibrium the mixture contains 0.120 mol of carbon monoxide. The total pressure of this mixture is 1.04×10^4 kPa. Calculate the partial pressure, in kPa, of hydrogen in the equilibrium mixture.

[4 marks]

05.2 Give an expression for the equilibrium constant (K_p) for this reaction. State the units.

[2 marks]

05.3 Some more carbon monoxide is added to the mixture. The new mixture is allowed to reach equilibrium at temperature T. State the effect, if any, on the partial pressure of methanol and on the value of K_p.

[2 marks]

05.4 State the effect, if any, of the addition of a catalyst on the value of K_p for this equilibrium. Explain your answer.

[2 marks]

06 Compounds A and B react together to form an equilibrium mixture containing compounds C and D according to the equation

$2A + B \rightleftharpoons 3C + D$

06.1 A beaker contained 40 cm^3 of a 0.16 mol dm^{-3} aqueous solution of A. 9.5×10^{-3} mol of B and 2.8×10^{-2} mol of C were added to the beaker and the mixture was left to reach equilibrium. The equilibrium mixture formed contained 3.9×10^{-3} mol of A. Calculate the amounts, in moles, of B, C and D in the equilibrium mixture.

[5 marks]

06.2 Give the expression for the equilibrium constant (K_c) for this equilibrium and its units.

[2 marks]

06.3 A different equilibrium mixture of these four compounds, at a different temperature, contained 0.21 mol of B, 1.05 mol of C and 0.076 mol of D in a total volume of 5.00×10^2 cm^3 of solution. At this temperature the numerical value of K_c was 116. Calculate the concentration of A, in mol dm^{-3}, in this equilibrium mixture. Give your answer to the appropriate number of significant figures.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Jun 2019 Q12*	1
02	No effect	*Source: Practice question (written for this set)*	1
03	**D**	*Source: 7405/3 Jun 2023 Q13*	1
04	**B**	*Source: 7404/2 Jun 2022 Q18*	1
05.1	M1 equilibrium n(H_2) = 0.24 M2 total moles = 0.47 M3 mole fraction H_2 = 0.24 ÷ 0.47 = 0.51(1) M4 p(H_2) = 5310 or 5.31 × 10^3 kPa	*Source: 7405/1 Nov 2020 Q06.2–06.5*	4
05.2	M1 K_p = p(CH_3OH) ÷ (p(H_2)^2 × p(CO)) M2 kPa^{-2} or Pa^{-2}	Do not allow square brackets	2
05.3	M1 partial pressure of methanol increases M2 K_p no effect		2
05.4	M1 no effect M2 increases rate of forward and backward reactions equally / catalyst does not affect position of equilibrium	M2 allow only temperature affects K_p	2
06.1	M1 initial n(A) = 6.4 × 10^{-3} M2 6.4 × 10^{-3} – 2x = 3.9 × 10^{-3} so x = 1.25 × 10^{-3} M3 B = 9.5 × 10^{-3} – x = 8.25 × 10^{-3} M4 C = 2.8 × 10^{-2} + 3x = 0.0318 M5 D = x = 1.25 × 10^{-3}	*Source: 7405/2 Jun 2018 Q04.1–04.3* If M1 wrong can score max 3 Allow 2 or more sig figs	5
06.2	K_c = $[C]^3[D] \div [A]^2[B]$ mol dm^{-3}	If K_c wrong, no mark for units	2
06.3	M1 $[A]^2$ = $[C]^3[D] \div K_c[B]$ M2 division of mol of B, C and D by correct volume ($[A]^2$ = 0.0289 or 0.0290) M3 [A] = 0.17 (must be 2 sf)		3